

Problem 9.3**Solution:****Definition**

A *perfect loop* is a set of three disjoint circular intervals on the real line that join end-to-end in a cycle.

$$[k + 1, \infty) \cup (-\infty, i], [i + 1, j], [j + 1, k], \quad i, j, k \in \mathbb{R}^+$$

We will show a bijection between directed triangles in the graph and such perfect loops in the interval instance.

Interval Construction

Let $M = n + 1$. For each directed edge $(i \rightarrow j) \in E$, construct three circular intervals:

$$\begin{aligned} I_{ij}^{(1)} &= [i, M + j - 1], \\ I_{ij}^{(2)} &= [M + i, 2M + j - 1], \\ I_{ij}^{(3)} &= [2M + i, \infty) \cup (-\infty, j - 1]. \end{aligned}$$

By construction, for any triangle $(i \rightarrow j \rightarrow k \rightarrow i)$, the three intervals

$$I_{ij}^{(1)}, \quad I_{jk}^{(2)}, \quad I_{ki}^{(3)}$$

are exactly a perfect loop as defined above. (Similarly for $I_{ij}^{(2)}, I_{jk}^{(3)}, I_{ki}^{(1)}$ and $I_{ij}^{(3)}, I_{jk}^{(1)}, I_{ki}^{(2)}$.)

Since $M > n$, any two intervals of the same type (i.e., both of the form $I^{(1)}$, $I^{(2)}$, or $I^{(3)}$) will necessarily overlap. Therefore, any disjoint subset of intervals can contain at most one interval of each type, implying that the maximum size of a disjoint subset is 3, and any such subset of size 3 must consist of exactly one interval of each type.

Weight Assignment

Let $W = 3 \cdot \max_{(i \rightarrow j) \in E} w_{ij}$. Assign

$$W_{ij}^* = (M + j - i)W - w_{ij}.$$

Since $w_{ij} > 0$ and $M = n + 1$, we have $W_{ij}^* > 0$.

For any disjoint set of size 3, its total weight is

$$(3M + v' - v + u' - u + w' - w)W - (w_{vv'} + w_{uu'} + w_{ww'}),$$

where

$$w' < v < v' < u < u' < w, \quad u, u', v, v', w, w' \in \mathbb{R}^+.$$

Note that $v' - v + u' - u + w' - w \leq 0$, $W > (w_{vv'} + w_{uu'} + w_{ww'})$. From this, we can conclude that for any set of three disjoint circular intervals the total weight is always at most $3MW$.

For a perfect loop, since it corresponds to a triangle, we have $v' = u$, $u' = w$, $w' = v$ thus $v' - v + u' - u + w' - w = 0$, its total weight is at least $(3M - 1)W$. Meanwhile, for any set of three disjoint intervals that does not form a perfect loop, we have $v' - v + u' - u + w' - w < 0$, its total weight strictly less than $(3M - 1)W$.

Correctness of the Reduction

($\Rightarrow$) If $(i \rightarrow j \rightarrow k \rightarrow i)$ is a directed triangle, then selecting $I_{ij}^{(1)}, I_{jk}^{(2)}, I_{ki}^{(3)}$ yields three disjoint intervals forming a perfect loop. Their total weight is

$$3MW - (w_{ij} + w_{jk} + w_{ki}).$$

where $(w_{ij} + w_{jk} + w_{ki})$ is the weight of the triangle. Since any set of three disjoint intervals that does not form a perfect loop must have smaller weight, the MIN-WEIGHT-TRIANGLE corresponds to the max-weight CIRCULAR-INTERVAL-SELECTION.

($\Leftarrow$) Conversely, any set of three disjoint intervals that does not form a perfect circular loop will either contain overlapping intervals (and thus be invalid) or have a total weight strictly less than that of any perfect loop. Therefore, the disjoint subset with maximum total weight must consist of exactly three intervals that form a perfect circular loop. By the interval construction, such a perfect loop corresponds precisely to a directed triangle $(u \rightarrow v \rightarrow w \rightarrow u)$ in the graph G .

Complexity

This reduction runs in $O(|E|)$ time and produces $N = 3|E| = O(n^2)$ intervals. An $O(N^{1.499})$ algorithm for CIRCULAR-INTERVAL-SELECTION would yield an $O(n^{2.999})$ algorithm for MIN-WEIGHT-TRIANGLE, contradicting the conjectured hardness.